

THE BURKHOLDER MARTINGALE TRANSFORM INEQUALITY IN LEAN

1. OVERVIEW

This repository proves the *Burkholder inequality for the real-valued martingale transforms* in the Lean proof assistant, as stated in and in `MartingaleTransforms.lean` and Figure 1.

```
theorem Lp_Burkholder_inequality_martingaleTransform (p : ℝ≥0∞) (Ω : Type*)
  [mΩ : MeasurableSpace Ω] (μ : Measure Ω)
  [IsFiniteMeasure μ] (ℱ : Filtration ℕ mΩ) (w f : ℕ → Ω → ℝ)
  (hp_one : 1 < p)
  (hfin: p ≠ ∞)
  (hstrong : IsStronglyPredictable ℱ w)
  (hmart : Martingale f ℱ μ)
  (hLp : ∀ n, MemLp (f n) p μ)
  (hbound : ∀ n, ∀ ω ∂μ, |w n ω| ≤ 1):
  ∀ n, eLpNorm ((w *m f) n) p μ ≤
    ENNReal.ofReal (Burkholder.pStar p.toReal - 1) * eLpNorm (f n) p μ
```

FIGURE 1.

That is the **Burkholder inequality for the real-valued martingale transforms**

Theorem 1 (Burkholder martingale transform inequality). *Let $p \in (1, \infty)$. Let $(\Omega, \mathcal{A})$ be a measurable space, and let μ be a finite measure on Ω . Let $\mathcal{F} = (\mathcal{F}_n)_{n \in \mathbb{N}}$ be a filtration indexed by $\mathbb{N}$, and let*

$$w_n, f_n : \Omega \rightarrow \mathbb{R}.$$

be measurable functions such that $f = (f_n)_{n \in \mathbb{N}}$ is a discrete martingale with respect to $\mathcal{F}$. Moreover

$$f_n \in L^p(\mu) \quad \text{for every } n \in \mathbb{N}.$$

Suppose that $w = (w_n)_{n \in \mathbb{N}}$ predictable sequence with respect to $\mathcal{F}$ with

$$|w_n(\omega)| \leq 1 \quad \text{for every } n \in \mathbb{N} \text{ and for } \mu\text{-a.e. } \omega.$$

Then, for every $n \in \mathbb{N}$,

$$\|(w \star_m f)_n\|_{L^p(\mu)} \leq (p^* - 1) \|f_n\|_{L^p(\mu)},$$

where $w \star_m f$ is the martingale transform of f with respect to w and $p^ = \max\{p, (p-1)/p\}$.*

See Bañuelos and Davis 2011 [1] for more information on this interesting result.

2. THE BURKHOLDER FUNCTION v

The proof follows Burkholder 1985 [2]. The main step there and the analytic heart of the formalization is to prove that the function

$$v(p, x, y) = \left| \frac{x+y}{2} \right|^p - |p^* - 1|^p \left| \frac{x-y}{2} \right|^p.$$

has a majorant u satisfying the theorem in Figure 2 and as stated in `Majorant.lean`. That is,

Theorem 2. *Let $p \in \mathbb{R}$ with $p > 1$. Then there exist*

$$u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} : \mathbb{R} \rightarrow \mathbb{R}$$

and a constant $C \geq 0$ such that

```

theorem exists_majorant_p_g_1 (p : ℝ) (hp : p > 1) :
  ∃ u du_dx du_dy : ℝ → ℝ → ℝ, ∃ C : ℝ,
    0 ≤ C ∧
    ContinuousOn (fun z : ℝ × ℝ => u z.1 z.2) Set.univ ∧
    ContinuousOn (fun z : ℝ × ℝ => du_dx z.1 z.2) Set.univ ∧
    ContinuousOn (fun z : ℝ × ℝ => du_dy z.1 z.2) Set.univ ∧
    (∀ x y,
      |u x y| ≤ C * (Real.rpow |x| p + Real.rpow |y| p)) ∧
    (∀ x y,
      |du_dx x y| ≤ C * (Real.rpow |x| (p - 1) + Real.rpow |y| (p - 1))) ∧
    (∀ x y,
      |du_dy x y| ≤ C * (Real.rpow |x| (p - 1) + Real.rpow |y| (p - 1))) ∧
    (∀ x y h k, h * k ≤ 0 →
      u (x + h) (y + k) ≤ u x y + du_dx x y * h + du_dy x y * k) ∧
    (∀ x y, v p x y ≤ u x y) ∧
    (∀ x y, x * y ≤ 0 → u x y ≤ 0) ∧
    (∀ x y, p ≠ 2 ∧ x * y = 0 ∧ (x, y) ≠ (0, 0) → u x y < 0) := by

```

FIGURE 2.

A. The functions

$$(x, y) \mapsto u(x, y), \quad (x, y) \mapsto \frac{\partial u}{\partial x}(x, y), \quad (x, y) \mapsto \frac{\partial u}{\partial y}(x, y)$$

are continuous on $\mathbb{R}^2$,

B. For all $x, y, h, k \in \mathbb{R}$,

$$\begin{aligned}
 |u(x, y)| &\leq C(|x|^p + |y|^p), \\
 \left| \frac{\partial u}{\partial x}(x, y) \right| &\leq C(|x|^{p-1} + |y|^{p-1}), \\
 \left| \frac{\partial u}{\partial y}(x, y) \right| &\leq C(|x|^{p-1} + |y|^{p-1}),
 \end{aligned}$$

C. We have

$$hk \leq 0 \implies u(x + h, y + k) \leq u(x, y) + \frac{\partial u}{\partial x}(x, y)h + \frac{\partial u}{\partial y}(x, y)k,$$

D. We have

$$v(p, x, y) \leq u(x, y),$$

E. We have

$$xy \leq 0 \implies u(x, y) \leq 0,$$

and

F. We have

$$p \neq 2, \quad xy = 0, \quad (x, y) \neq (0, 0) \implies u(x, y) < 0.$$

The proof of Theorem 2 is particularly painful to formalize in Lean. The function u is given explicitly in [2], but it is defined piecewise on several sectors of $\mathbb{R}^2$. Although the argument used to prove Theorem 2 in [2] takes only about half a page, is fairly simple, and could be followed by an undergraduate student with a solid background in point-set topology and analysis, a fully formalization would probably require several weeks of work (as Codex itself suggested to me), and perhaps even months.

In our case, the formalization became feasible within a manageable amount of time, namely a few days, by using AI tools such as Copilot and Codex in *agent mode*. In practice, we barely had to type any Lean code ourselves; instead, we mostly directed the agent when needed. Unfortunately, even with the paid version of Codex — through ChatGPT Plus, together with the Codex extension for Visual Studio Code — we frequently ran out of credits every day, and had to spend time purchasing additional credits and restarting sessions in different computing environments.

The result was more than 20,000 lines of Lean code. We believe that this code is unlikely to be optimal; nevertheless, it is remarkable that we barely had to write any proof by hand. Most of the work consisted in guiding the agent using colloquial language and image snapshots from [2]. This is quite astonishing when compared with the capabilities of pre-agent AI tools only a few months ago.

REFERENCES

- [1] Rodrigo Bañuelos and Burgess Davis. *Donald Burkholder's Work in Martingales and Analysis*, pages 1–22. Springer New York, New York, NY, 2011.
- [2] D. L. Burkholder. An elementary proof of an inequality of R. E. A. C. Paley. *Bull. Lond. Math. Soc.*, 17:474–478, 1985.